

CURRICULUM VITAE

Chongzhi Zang, PhD

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EDUCATION AND TRAINING

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|---|---|-----------|
| • Peking University, Beijing, China | B.S., Physics | 07/2005 |
| • The George Washington University | Ph.D., Physics | 09/2010 |
| • Harvard University/Dana-Farber Cancer Institute | Postdoctoral training,
Computational Biology | 2010–2016 |
| • Leadership in Academic Matters (LAM) Program,
University of Virginia | Leadership training | 2024–2025 |
| • UVA Health Leadership Institute (HLI) | Leadership training | 2025–2026 |

APPOINTMENTS

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|-------------------|--|
| 11/2016 – | University of Virginia , Charlottesville, VA
<i>Professor of Genome Sciences</i> (07/2026–)
<i>Professor of Biochemistry and Molecular Genetics</i> (by Courtesy) (07/2026–)
<i>Professor of Biomedical Engineering</i> (by Courtesy) (07/2026–)
<i>Director of Computational Genomics, Cancer Center</i> (07/2024–)
<i>Associate Professor of Genome Sciences</i> (07/2024–06/2026)
<i>Associate Professor of Biochemistry and Molecular Genetics</i> (07/2022–06/2026)
<i>Associate Professor of Biomedical Engineering</i> (07/2022–06/2026)
<i>Associate Professor of Public Health Sciences</i> (07/2022–06/2024)
<i>Resident Faculty, Center for Public Health Genomics</i> (11/2016–07/2024)
<i>Full Member, UVA Comprehensive Cancer Center</i> (01/2017–)
<i>Assistant Professor of Public Health Sciences, Biochemistry and Molecular Genetics, and Biomedical Engineering</i> (11/2016–06/2022) |
| 07/2010 – 10/2016 | Harvard University/Dana-Farber Cancer Institute , Boston, MA
<i>Postdoctoral Fellow</i> , Department of Data Science, Dana-Farber Cancer Institute;
and Department of Biostatistics, Harvard T.H. Chan School of Public Health
(Postdoctoral Mentor: Xiaole Shirley Liu, PhD) |
| 05/2007 – 06/2010 | National Institutes of Health , Bethesda, MD
<i>Predocctoral Research Assistant</i> , Laboratory of Epigenome Biology,
National Heart, Lung, and Blood Institute
(PhD Co-advisor: Keji Zhao, PhD) |

- 09/2005 – 06/2010 **The George Washington University**, Washington, DC
Graduate Research Assistant, Department of Physics (09/2006–06/2010)
Graduate Teaching Assistant, Department of Physics (09/2005–05/2008)
 (PhD Advisor: Weiqun Peng, PhD)
- 02/2004 – 07/2005 **Institute of Physics, Chinese Academy of Sciences**, Beijing, China
Undergraduate Research Assistant, High-field Physics Group (L05)
 (Thesis Advisor: Jie Zhang, PhD)

PUBLICATIONS

Total citations > 22,000; H-index = 46 (Google Scholar)

*: equal-contributing authors; §: co-corresponding authors. Underscored: Dr. Zang's trainees.

Preprints Under Peer Review

1. Hongpan Zhang§, Bhummanat Sittipongpittaya, **Chongzhi Zang**§. "GeneKnow: An auditable AI framework for source-grounded biological evidence synthesis." submitted. *bioRxiv* doi: <https://doi.org/10.64898/2026.05.28.728511> (2026).
2. Jingyi Wang, Hongpan Zhang, Zhenjia Wang, **Chongzhi Zang**. "BART-spatial unravels biologically significant transcriptional regulators from spatial omics data." submitted. *bioRxiv* doi: <https://doi.org/10.64898/2026.05.05.723027> (2026).
3. Hongpan Zhang, Lei Kang, Jingyi Wang, Kathleen P. Liang, Zhenjia Wang, Kexin Xu, **Chongzhi Zang**. "BARTsc identifies key transcriptional regulators from single-cell omics data." under revision. *bioRxiv* doi: <https://doi.org/10.64898/2026.02.24.707729> (2026).
4. Tara Akhshi*, Shengen Shawn Hu*, Esme Wheeler, Christian Hellriegel, Douglas S. Richardson, Nicole Cayting, Nicole A. Traphagen, Wangu Mvula, Buraq Ahmed, Rinath Jeselsohn, **Chongzhi Zang**§, Myles Brown§. "Super-resolution microscopy reveals distinct epigenetic states regulated by estrogen receptor activity." accepted to *Nature Communications*. Preprint at *bioRxiv* doi: <https://doi.org/10.1101/2025.06.16.659976> (2025).
5. Shengyuan Wang, Zhongyang Wu, Zhe Zhong, Xu Zhou§, **Chongzhi Zang**§. "Transcriptional condensates at super-enhancers mediate pH-dependent transcriptional control in innate immunity." submitted. *bioRxiv* doi: <https://doi.org/10.1101/2025.10.29.685293> (2025).

Peer-reviewed Journal Articles

6. Zhaoxia Ma, Miao Liu, Shengyuan Wang, Siyuan Wang§, **Chongzhi Zang**§. "Differential analysis of image-based chromatin tracing data with Dory." *Genome Biology*. doi: <https://doi.org/10.1186/s13059-026-04264-y> (2026).
7. Wei Hu*, Qing Chen*, Shaoqi Zhu*, Shengen Shawn Hu*, Haibo Yu, Vanita Patel, Ying Wang, Vladimir P. Badovinac, Yi Zhang, **Chongzhi Zang**§, Weiqun Peng§, Hai-Hui Xue§. "Mapping self-associating chromatin hubs identifies Id proteins as key determinants of exhausted CD8+ T cell

fate.” **Nature Immunology** 27, 1678–1692 (2026).

8. Yiran Wang*, Jindayi Liang*, Kimberly S. Lange, Justin Demmerle, Eleanor A. Liu, Ethan Black, Britney Jiayu He, Christopher J. Ricketts, Shawn R. Yoshida, Shasha Chong, W. Marston Linehan, Jennifer M. Kavan, **Chongzhi Zang**, Danfeng Cai. “TEAD1 condensates are transcriptionally inactive storage sites on the pericentromeric heterochromatin in cancer cells.” **Nature Cell Biology** 28, 1480–1495 (2026).
9. Shengen Shawn Hu, Zhangli Su, Lin Liu, Qingying Chen, Megan C. Grieco, Mengxue Tian, Anindya Dutta, **Chongzhi Zang**. “PATTY corrects open-chromatin bias for improved bulk and single-cell CUT&Tag profiling.” **Nature Communications** 17, 6710 (2026).
10. Subhash C. Prajapati, Cem Meydan, Yaseswini Neelamraju, Zhenjia Wang, Grant Atwood, B. Bishal Paudel, Hao Fan, Nicholas Dunham, Richard Dillon, Jorge A Gandara, Tak Lee, Caroline Sheridan, Shengyuan Wang, Paul Zumbo, Michael W. Becker, Lars Bullinger, Martin P Carroll, Richard J D’Andrea, Ross L Levine, Christopher E. Mason, Ari M Melnick, **Chongzhi Zang**, Stefan Bekiranov, Francine E. Garrett-Bakelman. “CCAAT-enhancer binding protein delta functions as a tumor suppressor gene in acute myeloid leukemia.” **Neoplasia** 74, 101293 (2026).
11. Yue Wu, Chaofan Li, Jinyi Tang, Xiaochen Gao, In Su Cheon, Bibo Zhu, Ruixuan Zhang, Cori E Fain, Shengen Shawn Hu, Harish Narasimhan, Gislane de Almeida Santos, Katayoun Ayasoufi, Aaron J Johnson, Hui Zong, **Chongzhi Zang**, Haidong Dong, Jie Sun. “Temporal single-cell analysis reveals age-associated delay in immune resolution after respiratory viral infection.” **eLife** 15, RP109988 (2026).
12. Vinay Singh Tanwar*, Vajir Malek*, Jingyi Wang*, Yingjun Luo, Naseeb Kaur Malhi, Hongpan Zhang, Maryam Abdollahi, Linda Lanting, Parijat Senapati, Sadhan Das, Marpadga A. Reddy, **Chongzhi Zang**, Clint L. Miller[§], Zhen Bouman Chen[§], Rama Natarajan[§]. “Single-cell multimodal profiling highlights persistent aortic smooth muscle cell changes in diabetic mice despite glycemic control.” **Arteriosclerosis, Thrombosis, and Vascular Biology** 46, e324012 (2026).
13. Nicholas Dunham*, Zhenjia Wang*, Yaseswini Neelamraju, Yan Guo, B. Bishal Paudel, Cem Meydan, Jorge A. Gandara, Fady Abdelmalak, Hao Fan, Joyce Hardwick, Justin H. Layer, Tak Lee, Bernhard Maier, W. Hayes McDonald, Isaani Patnaik, Subhash Prajapati, Franck Rapaport, Caroline Sheridan, Gloria Sheynkman, Paul Zumbo, Michael W. Becker, Lars Bullinger, Martin P. Carroll, Richard J. D’Andrea, Richard Dillon, Ross L. Levine, Christopher E. Mason, Ari M. Melnick, Donna S. Neuberg, Stefan Bekiranov[§], **Chongzhi Zang**[§], Utpal P. Dave[§], Francine E. Garrett-Bakelman[§]. “The LMO2-LDB1-TAL1 complex regulates transcription networks in Acute Myeloid Leukemia.” **Blood Neoplasia** 3, 100186 (2026).
14. Kevin Qiu, Qing Yin, **Chongzhi Zang**, Jianguo Tao. “Transcriptional condensates: epigenetic reprogramming and therapeutic targets in hematologic malignancies.” **Cancers** 17, 3148 (2025).
15. Xin Zhao*, Shengen Shawn Hu*, Wen-Han Lee, Johannes L. Zakrzewski, Qing-Sheng Mi, Rachel K. Rosenstein, **Chongzhi Zang**[§], Xiaoke Ma[§], Hai-Hui Xue[§]. “Single-cell multiomics identifies Tcf1 and Lef1 as key initiators of early thymic progenitor fate.” **Science Immunology** 10, eadq8970 (2025).
16. Zhangli Su*, Mengxue Tian*, Etsuko Shibata, Yoshiyuki Shibata, Tianyi Yang, Zhenjia Wang, Fulai Jin, **Chongzhi Zang**[§], Anindya Dutta[§]. “Regulation of epigenetics and chromosome structure by human ORC2.” **Cell Reports** 44 (6), 115816 (2025).

17. Wei Hu, Shengen Shawn Hu, Shaoqi Zhu, Weiqun Peng, Vladimir P. Badovinac, **Chongzhi Zang**, Xudong Zhao[§], Hai-Hui Xue[§]. "Hdac1 as an early determinant of intermediate-exhausted CD8⁺ T cell fate in chronic viral infection." ***Proceedings of the National Academy of Sciences USA*** 122 (19), e2502256122 (2025).
18. Yubao Cheng, Mengwei Hu, Bing Yang, Tyler B. Jensen, Yuan Zhang, Tianqi Yang, Ruihuan Yu, Zhaoxia Ma, Jonathan S. D. Radda, Shengyan Jin, **Chongzhi Zang**, Siyuan Wang. "Perturb-tracing enables high-content screening of multi-scale 3D genome regulators." ***Nature Methods*** 22, 950–961 (2025).
19. Shengyuan Wang^{*}, Zhenjia Wang^{*}, **Chongzhi Zang**. "Genomic clustering tendency of transcription factors reflects phase-separated transcriptional condensates at super-enhancers." ***Nucleic Acids Research*** 53, gkaf015 (2025).
20. Shengen Shawn Hu, Hai-Hui Xue, **Chongzhi Zang**. "IGN: Invariable gene set-based normalization for chromatin accessibility profile data analysis." ***Computational and Structural Biotechnology Journal*** 27, 501–507 (2025).
21. Harish Narasimhan^{*}, In Su Cheon^{*}, Wei Qian^{*}, Shengen Shawn Hu^{*}, Tanyalak Parimon^{*}, Chaofan Li, Nick Goplen, Yue Wu, Xiaoqin Wei, Young Min Son, Elizabeth Fink, Gislane de Almeida Santos, Jinyi Tang, Changfu Yao, Lyndsey Muehling, Glenda Canderan, Alexandra Kadl, Abigail Cannon, Samuel Young, Riley Hannan, Grace Bingham, Mohammed Arish, Arka Sen Chaudhari, Jun sub Im, Cameron L. R. Mattingly, Patcharin Pramoonjago, Alberto Marchesvsky, Jeffrey Sturek, Jacob E. Kohlmeier, Yun Michael Shim, Judith Woodfolk, **Chongzhi Zang**[§], Peter Chen[§], Jie Sun[§]. "An aberrant immune–epithelial progenitor niche drives viral lung sequelae." ***Nature*** 634, 961–969 (2024).
22. Yen T.M. Nguyen, Lydia Sibley^{*}, Piotr Przanowski^{*}, Xiao-Yu Zhao^{*}, Michael Kovacs, Shengyuan Wang, Marieke K. Jones, Maureen Cowan, Wenjie Liu, Andrea R. Merchak, Alban Gaultier, Kevin Janes, **Chongzhi Zang**, Tajie Harris, Sarah E. Ewald[§], Hui Zong[§]. "*Toxoplasma gondii* infection supports the infiltration of T cells into brain tumors." ***Journal of Neuroimmunology*** 393, 578402 (2024).
23. Nikolai G. Vetr[§], Nicole R. Gay, MoTrPAC Study Group (including Benjamin G. Ke and **Chongzhi Zang**), Stephen B. Montgomery[§]. The impact of exercise on gene regulation in association with complex trait genetics. ***Nature Communications*** 15, 3346 (2024).
24. The MoTrPAC Study Group (including Benjamin G. Ke and **Chongzhi Zang**). "Temporal dynamics of the multi-omic response to endurance exercise training." ***Nature*** 629, 174–183 (2024).
25. Mengxue Tian, Zhenjia Wang, Zhangli Su, Etsuko Shibata, Yoshiyuki Shibata, Anindya Dutta[§], **Chongzhi Zang**[§]. "Integrative analysis of DNA replication origins and ORC-/MCM-binding sites in human cells reveals a lack of overlap." ***eLife*** 12, RP89548 (2024).
26. Heather M. Raimer Young, Pei-Chi Hou, Anna R. Bartosik, Naomi D. Atkin, Lixin Wang, Zhenjia Wang, Aakrosh Ratan, **Chongzhi Zang**, Yuh-Hwa Wang. "DNA fragility at topologically associated domain boundaries is promoted by alternative DNA secondary structure and topoisomerase II activity." ***Nucleic Acids Research***, gkae164 (2024).
27. Xin Zhao, Wei Hu, Sung Rye Park, Shaoqi Zhu, Shengen Shawn Hu, **Chongzhi Zang**, Weiqun Peng, Qiang Shan[§], Hai-Hui Xue[§]. "The transcriptional cofactor Tle3 reciprocally controls effector

and central memory CD8⁺ T cell fates.” **Nature Immunology** 25, 294–306 (2024).

28. Hyun Jae Cho, Zhenjia Wang, Yidan Cong, Stefan Bekiranov, Aidong Zhang[§], **Chongzhi Zang**[§]. “DARDN: a deep-learning approach for CTCF binding sequence classification and oncogenic regulatory feature discovery.” **Genes** 15(2), 144 (2024).
29. Jose Verdezoto Mosquera, Gaëlle Auguste, Doris Wong, Adam W. Turner, Chani J. Hodonsky, Astrid Catalina Alvarez-Yela, Yipei Song, Qi Cheng, Christian L. Lino Cardenas, Konstantinos Theofilatos, Maxime Bos, Maryam Kavousi, Patricia A. Peyser, Manuel Mayr, Jason C. Kovacic, Johan L. M. Björkegren, Rajeev Malhotra, P. Todd Stukenberg, Alope V. Finn, Sander W. van der Laan, **Chongzhi Zang**, Nathan C. Sheffield, Clint L. Miller. “Integrative single-cell meta-analysis reveals disease-relevant vascular cell states and markers in human atherosclerosis.” **Cell Reports** 42, 113380 (2023).
30. Yue Wu, Shengen Shawn Hu, Ruixuan Zhang, Nick P Goplen, Xiaochen Gao, Harish Narasimhan, Ao Shi, Yin Chen, Ying Li, **Chongzhi Zang**, Haidong Dong, Thomas J Braciale, Bibo Zhu[§], Jie Sun[§]. “Single cell RNA sequencing unravels mechanisms underlying senescence-like phenotypes of Alveolar Macrophages.” **iScience** 26, 107197 (2023).
31. Manikarna Dinda, Ryan D. Fine, Shekhar Saha, Zhenjia Wang, **Chongzhi Zang**, Mingguang Li, Jeffrey S. Smith. “Fob1-dependent condensin recruitment and loop extrusion on yeast chromosome III.” **PLOS Genetics** 19(4), e1010705 (2023).
32. Hyun Jae Cho, Mia Shu, Stefan Bekiranov, **Chongzhi Zang**, Aidong Zhang. “Interpretable meta-learning of multi-omics data for survival analysis and pathway enrichment.” **Bioinformatics**, btad113 (2023).
33. Shengen Shawn Hu, Lin Liu, Qi Li, Wenjing Ma, Michael J. Guertin, Clifford A. Meyer, Ke Deng, Tingting Zhang, **Chongzhi Zang**. “Intrinsic bias estimation for improved analysis of bulk and single-cell chromatin accessibility profiles using SELMA.” **Nature Communications** 13, 5533 (2022).
34. Yuchao Jiang[§], Yuriko Harigaya, Zhaojun Zhang, Hongpan Zhang, **Chongzhi Zang**, Nancy R. Zhang[§]. “Nonparametric single-cell multiomic characterization of trio relationships between transcription factors, target genes, and cis-regulatory regions.” **Cell Systems** 13, 737–751 (2022).
35. William F. Lamberti, **Chongzhi Zang**. “Extracting physical characteristics of higher-order chromatin structures from 3D image data.” **Computational and Structural Biotechnology Journal** 20, 3387–3398 (2022).
36. Ryan J. Smith*, Hongpan Zhang*, Shengen Shawn Hu*, Theodora Yung, Roshane Francis, Lilian Lee, Mark W. Onaitis, Peter B. Dirks, **Chongzhi Zang**[§], Tae-Hee Kim[§]. “Single-cell chromatin profiling of the primitive gut tube reveals regulatory dynamics underlying lineage fate decisions.” **Nature Communications** 13, 2965 (2022).
37. Adam W. Turner*, Shengen Shawn Hu*, Jose Verdezoto Mosquera, Wei Feng Ma, Chani J. Hodonsky, Doris Wong, Gaëlle Auguste, Yipei Song, Katia Sol-Church, Emily Farber, Soumya Kundu, Anshul Kundaje, Nicolas G. Lopez, Lijiang Ma, Saikat Kumar B, Ghosh, Suna Onengut-Gumuscu, Euan A. Ashley, Thomas Quertermous, Alope V. Finn, Nicholas J. Leeper, Jason C. Kovacic, Johan L. M. Björkegren, **Chongzhi Zang**[§], Clint L. Miller[§]. “Single-nucleus chromatin accessibility profiling highlights regulatory mechanisms of coronary artery disease risk.” **Nature Genetics** 54, 804–816 (2022).

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38. Qiang Shan*, Shengen Shawn Hu*, Shaoqi Zhu, Xia Chen, Vladimir P. Badovinac, Weiqun Peng, **Chongzhi Zang**[§], Hai-Hui Xue[§]. “Tcf1 preprograms the mobilization of glycolysis in central memory CD8⁺ T cells during recall responses.” **Nature Immunology** 23, 386–398 (2022).
39. Avik Dutta, Dipmoy Nath, Yue Yang, Bao T. Le, Mohammad Ferdous-Ur Rahman, Patrick Faughnan, Zhenjia Wang, Matthew Stuver, Rongquan He, Wuwei Tan, Robert E. Hutchison, Jason M. Foulks, Steven L. Warner, **Chongzhi Zang**, Golam Mohi. “Genetic ablation of Pim1 or pharmacologic inhibition with TP-3654 ameliorates myelofibrosis in murine models.” **Leukemia** 36, 746–759 (2022).
40. Bi Shi*, Wei Li*, Yansu Song*, Zhenjia Wang*, Rui Ju, Aleksandra Ulman, Jing Hu, Francesco Palomba, Yanfang Zhao, John Philip Le, William Jarrard, David Dimoff, Michelle A. Digman, Enrico Gratton, **Chongzhi Zang**, Hao Jiang. “UTX condensation underlies its tumor suppressive activity.” **Nature** 597, 726–731 (2021).
41. Naomi D. Atkin, Heather M. Raimier, Zhenjia Wang, **Chongzhi Zang**, Yuh-Hwa Wang. “Assessing acute myeloid leukemia susceptibility in rearrangement-driven patients by DNA breakage at topoisomerase II and CCCTC-binding factor/cohesin binding sites.” **Genes, Chromosomes and Cancer** 60, 808–821 (2021).
42. Zhenjia Wang, Yifang Zhang, **Chongzhi Zang**. “BART3D: Inferring transcriptional regulators from differential Hi-C data.” **Bioinformatics** 37, 3075–3078 (2021).
43. Wenjing Ma*, Zhenjia Wang*, Yifan Zhang, Neal E. Magee, Yayi Feng, Ruoyao Shi, Yang Chen, **Chongzhi Zang**. “BARTweb: a web server for transcriptional regulator association analysis.” **NAR Genomics and Bioinformatics** 3(2), lqab022 (2021).
44. Zachary V. Thomas, Zhenjia Wang, **Chongzhi Zang**. “BART Cancer: a web resource for transcriptional regulators in cancer genomes.” **NAR Cancer** 3, zcab011 (2021).
45. Avik Dutta, Yue Yang, Bao T. Le, Yifan Zhang, Omar Abdel-Wahab, **Chongzhi Zang**, Golam Mohi. “U2af1 is critical for survival and function of hematopoietic stem/progenitor cells.” **Leukemia** 35, 2382–2398 (2021).
46. **Chongzhi Zang**[§], Yiren Wang, Weiqun Peng[§]. “RECOGNICER: a coarse-graining approach for identifying broad domains from ChIP-seq data.” **Quantitative Biology** 8, 359–368 (2020).
47. Celestia Fang*, Zhenjia Wang*, Cuijuan Han, Stephanie L Safgren, Kathryn A Helmin, Emmalee R Adelman, Kyle P Eagen, Alexandre Gaspar-Maia, Maria E Figueroa, Benjamin D Singer, Aakrosh Ratan, Panagiotis Ntziachristos[§], **Chongzhi Zang**[§]. “Cancer-specific CTCF binding facilitates oncogenic transcriptional dysregulation.” **Genome Biology** 21, 247 (2020).
48. Danqi Chen*, Qiao Yi Chen*, Zhenjia Wang*, Yusha Zhu, Thomas Kluz, Wuwei Tan, Jinqian Li, Feng Wu, Lei Fang, Xiaoru Zhang, Rongquan He, Steven Shen, Hong Sun, **Chongzhi Zang**[§], Chunyuan Jin[§], Max Costa[§]. “Polyadenylation of histone H3.1 mRNA promotes cell transformation by displacing H3.3 from gene regulatory elements.” **iScience** 23, 101518 (2020).
49. The ENCODE Project Consortium (including **Chongzhi Zang**), Jill E. Moore*, Michael J. Purcaro*, Henry E. Pratt*, Charles B. Epstein*, Noam Shores*, Jessika Adrian*, Trupti Kawli*, Carrie A.

Davis*, Alexander Dobin*, Rajinder Kaul*, Jessica Halow*, Eric L. Van Nostrand*, Peter Freese*, David U. Gorkin*, Yin Shen*, Yupeng He*, Mark Mackiewicz*, Florencia Pauli-Behn*, Brian A. Williams, Ali Mortazavi, Cheryl A. Keller, Xiao-Ou Zhang, Shaimae I. Elhajjajy, Jack Huey, Diane E. Dickel, Valentina Snetkova, Xintao Wei, Xiaofeng Wang, Juan Carlos Rivera-Mulia, Joel Rozowsky, Jing Zhang, Surya B. Chhetri, Jialing Zhang, Alec Victorson, Kevin P. White, Axel Visel, Gene W. Yeo, Christopher B. Burge, Eric Lécuyer, David M. Gilbert, Job Dekker, John Rinn, Eric M. Mendenhall, Joseph R. Ecker, Manolis Kellis, Robert J. Klein, William S. Noble, Anshul Kundaje, Roderic Guigó, Peggy J. Farnham, J. Michael Cherry[§], Richard M. Myers[§], Bing Ren[§], Brenton R. Graveley[§], Mark B. Gerstein[§], Len A. Pennacchio[§], Michael P. Snyder[§], Bradley E. Bernstein[§], Barbara Wold[§], Ross C. Hardison[§], Thomas R. Gingeras[§], John A. Stamatoyannopoulos[§], Zhiping Weng[§]. “Expanded encyclopaedias of DNA elements in the human and mouse genomes.” *Nature* 583, 699–710 (2020).

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51. Jing Zhang*, Donghoon Lee*, Vineet Dhiman*, Peng Jiang*, Jie Xu*, Patrick McGillivray*, Hongbo Yang*, Jason Liu, William Meyerson, Declan Clarke, Mengting Gu, Shantao Li, Shaoke Lou, Jinrui Xu, Lucas Lochovsky, Matthew Ung, Lijia Ma, Shan Yu, Qin Cao, Arif Harmanci, Koon-Kiu Yan, Anurag Sethi, Gamze Gürsoy, Michael Rutenberg Schoenberg, Joel Rozowsky, Jonathan Warrell, Prashant Emani, Yucheng T. Yang, Timur Galeev, Xiangmeng Kong, Shuang Liu, Xiaotong Li, Jayanth Krishnan, Yanlin Feng, Juan Carlos Rivera-Mulia, Jessica Adrian, James R Broach, Michael Bolt, Jennifer Moran, Dominic Fitzgerald, Vishnu Dileep, Tingting Liu, Shenglin Mei, Takayo Sasaki, Claudia Trevilla-Garcia, Su Wang, Yanli Wang, **Chongzhi Zang**, Daifeng Wang, Robert J. Klein, Michael Snyder, David M. Gilbert, Kevin Yip, Chao Cheng, Feng Yue[§], X. Shirley Liu[§], Kevin White[§], Mark Gerstein[§]. “An integrative ENCODE resource for cancer genomics.” *Nature Communications* 11, 3696 (2020).
52. Qiang Shan*, Sheng'en Hu*, Xia Chen, Derek B. Danahy, Vladimir P. Badovinac, **Chongzhi Zang**[§], Hai-Hui Xue[§]. “Ectopic Tcf1 expression instills a stem-like program in exhausted CD8+ T cells to enhance viral and tumor immunity.” *Cellular & Molecular Immunology* 18, 1262–1277 (2021). doi:10.1038/s41423-020-0436-5 (2020).
53. Luke T. Oostdyk, Zhenjia Wang, **Chongzhi Zang**, Hui Li, Michael J. McConnell, Bryce M. Paschal. “An epilepsy-associated mutation in the nuclear import receptor KPNA7 reduces nuclear localization signal binding.” *Scientific Reports* 10, 4844 (2020).
54. Qingsu Cheng, Mina Khoshdeli, Bradley S. Ferguson, Kosar Jabbari, **Chongzhi Zang**[§], Bahram Parvin[§]. “YY1 is a cis-regulator in the organoid models of high mammographic density.” *Bioinformatics* 36, 1663–1667 (2020).
55. Cynthia C Jose*, Zhenjia Wang*, Vinay Singh Tanwar, Xiaoru Zhang, **Chongzhi Zang**[§], Suresh Cuddapah[§]. “Nickel induced transcriptional changes persist post exposure through epigenetic reprogramming.” *Epigenetics and Chromatin* 12, 75 (2019).
56. Teng Fei*, Wei Li*, Jingyu Peng*, Tengfei Xiao, Chen-Hao Chen, Alexander Wu, Jialiang Huang, **Chongzhi Zang**, X. Shirley Liu[§], Myles Brown[§]. “Deciphering essential cistromes using genome-

wide CRISPR screens.” *Proceedings of the National Academy of Sciences USA* 116, 25186–25195 (2019).

57. Kushani K. Shah, Robert H. Whitaker, Theodore Busby, Jing Hu, Bi Shi, Zhenjia Wang, **Chongzhi Zang**, William J. Placzek, Hao Jiang. “Specific inhibition of DPY30 activity by ASH2L-derived peptides suppresses blood cancer cell growth.” *Experimental Cell Research* 382(2), 111485 (2019).
58. Zhenjia Wang, Mete Civelek, Clint L. Miller, Nathan C. Sheffield, Michael J. Guertin, **Chongzhi Zang**. “BART: a transcription factor prediction tool with query gene sets or epigenomic profiles.” *Bioinformatics* 34, 2867–2869 (2018).
59. Tengfei Xiao*, Wei Li*, Xiaoqing Wang, Han Xu, Jixin Yang, Qiu Wu, Ying Huang, Joseph Geradts, Peng Jiang, Teng Fei, David Chi, **Chongzhi Zang**, Qi Liao, Jonathan Rennhack, Eran Andrechek, Nanlin Li, Simone Detre, Mitchell Dowsett, Rinath M. Jeselsohn, X. Shirley Liu[§], Myles Brown[§]. “Estrogen-regulated feedback loop limits the efficacy of estrogen receptor–targeted breast cancer therapy.” *Proceedings of the National Academy of Sciences USA* 115, 7869–7878 (2018).
60. Cynthia C Jose, Lakshmanan Jagannathan, Vinay Singh Tanwar, Xiaoru Zhang, **Chongzhi Zang**, Suresh Cuddapah. “Nickel exposure induces persistent mesenchymal phenotype in human lung epithelial cells through epigenetic activation of ZEB1.” *Molecular Carcinogenesis* 57, 794–806 (2018).
61. André L. Martins, Ninad M. Walavalkar, Warren D. Anderson, **Chongzhi Zang**, Michael J. Guertin. “Universal correction of enzymatic sequence bias reveals molecular signatures of protein/DNA interactions.” *Nucleic Acids Research* 46 (2), e9 (2017).
62. Shenglin Mei, Clifford A. Meyer, Rongbin Zheng, Qian Qin, Qiu Wu, Peng Jiang, Bo Li, Xiaohui Shi, Binbin Wang, Jingyu Fan, Celina Shih, Myles Brown, **Chongzhi Zang**[§], X. Shirley Liu[§]. “Cistrome Cancer: a web resource for integrative gene regulation modeling in cancer.” *Cancer Research* 77, e19–e22 (2017).
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Peer-reviewed Book Chapter

95. Qiuling Suo, Hyun Jae Cho, Jingyuan Chou, Stefan Bekiranov, **Chongzhi Zang**, Aidong Zhang. “Knowledge-guided meta learning for disease prediction.” In *Meta-Learning with Medical Imaging and Health Informatics Applications*. 253–273. Academic Press (2023).

Peer-reviewed Conference Papers/Abstracts

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97. Shengyuan Wang, Zhongyang Wu, Zhe Zhong, Zhenjia Wang, Xu Zhou, **Chongzhi Zang**. "Transcriptional condensates at super-enhancers mediate pH-dependent transcriptional control in innate immunity". **Cancer Research** 86 (7_Supplement): 1480. *AACR Annual Meeting Abstract 1480* (2026).
98. Hongpan Zhang, Jingyi Wang, Zhenjia Wang, **Chongzhi Zang**. "Motif-independent Identification of functional transcription factors from single-cell multiomics data." **Cancer Research** 85 (8_Supplement_1), 6294. *AACR Annual Meeting Abstract 6294* (2025).
99. Zhaoxia Ma, Miao Liu, Mengwei Hu, Shengyuan Wang, Siyuan Wang, **Chongzhi Zang**. "Dory: a computational method for differential chromatin tracing analysis." **Cancer Research** 85 (8_Supplement_1), 6308. *AACR Annual Meeting Abstract 6308* (2025).
100. Shengyuan Wang, Zhenjia Wang, Grant Atwood, Yanbin Xu, **Chongzhi Zang**. "CTCFDB: a systematic framework and comprehensive data resource for CTCF binding in the human genome." **Cancer Research** 85 (8_Supplement_1), 1074. *AACR Annual Meeting Abstract 1074* (2025).
101. Kevin Qiu, Jing Gao, Qing Yin, Allen Hu, Zhenjia Wang, Loryn Blackburn, Bijal Shah, Jeffrey W Craig, Craig Portell, Michael E Williams, **Chongzhi Zang**, Jianguo Tao. "Multi-omic analyses reveal epigenetic and transcriptomic reprogramming behind ibrutinib-venetoclax resistance development in mantle cell lymphoma." **Cancer Research** 85 (8_Supplement_1), 5364. *AACR Annual Meeting Abstract 5364* (2025).
102. Shengyuan Wang*, Zhenjia Wang*, **Chongzhi Zang**. "Clustered transcription factor binding reflects transcriptional condensates at cancer super-enhancers and associates with clinical outcome." **Cancer Research** 85 (5 Supplement), B045. *AACR Special Conference: Functional and Genomic Precision Medicine in Cancer* (2025).
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106. Shengyuan Wang, Zhenjia Wang, Yuh-Hwa Wang, Hui Li, P. Todd Stukenburg, **Chongzhi Zang**. "Integration analysis of lncRNAs and R-loops in 3D genome organization and gene regulation in the cancer genome." **Cancer Research** 84 (6 Supplement), 7355. *AACR Annual Meeting Abstract 7355* (2024).
107. Hongpan Zhang, Jingyi Wang, Zhenjia Wang, **Chongzhi Zang**. "BARTsc: A transcription factor analysis suite for single-cell omics data." **Cancer Research** 84 (6 Supplement), 2342. *AACR Annual Meeting Abstract 2342* (2024).

108. Zhenjia Wang, Shengyuan Wang, **Chongzhi Zang**. “Genomic clustering tendency of transcription factors reflects phase-separated transcriptional condensates at cancer super-enhancers.” **Cancer Research** 84 (6 Supplement), 1695. *AACR Annual Meeting Abstract* 1695 (2024).
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111. Mengxue Tian, Zhenjia Wang, Zhangli Su, Etsuko Shibata, Yoshiyuki Shibata, Anindya Dutta[§], **Chongzhi Zang**[§]. “Integrative analysis of genome-wide DNA replication origins in human cells.” **Cancer Research** 83 (7 Supplement), 3174. *AACR Annual Meeting Abstract* 3174 (2023).
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 121. Hongfang Wang*, **Chongzhi Zang***, Len Taing, Hoifung Wong, Yumi Yashiro-Ohtani, Stephen Blacklow, Warren S. Pear, X. Shirley Liu, Jon C. Aster. "Genome-wide analysis of NOTCH1, ETS family factors, and RUNX1 binding in human T lymphoblastic leukemia cells reveals distinct regulatory elements." *Blood* 120 (21), *ASH Annual Meeting Abstracts* 1277 (2012).
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SOFTWARE AND RESOURCES

1. SICER: Spatial-clustering approach for the Identification of ChIP-Enriched Regions, a ChIP-Seq data analysis method and software package. <http://home.gwu.edu/~wpeng/Software.htm>
2. RECOGNICER: Recursive Coarse-Graining approach for the Identification of ChIP-Enriched Regions, a ChIP-Seq ultra-broad peak calling method. <https://github.com/zanglab/recognicer>

3. MANCIE: Matrix Analysis and Normalization by Concordant Information Enhancement, a computational method for high-dimensional genomic data integration. <https://cran.r-project.org/web/packages/MANCIE/>
4. MARGE: Model-based Analysis of Regulation of Gene Expression, a comprehensive computational method for inference of cis-regulation of gene expression leveraging public H3K27ac genomic profiles in human or mouse. <http://cistrome.org/MARGE/>
5. BART: Binding Analysis for Regulation of Transcription, a bioinformatics tool for predicting functional transcription factors that bind at genomic cis-regulatory regions to regulate gene expression in the human or mouse genomes, given a query gene set or a ChIP-seq dataset as input. <https://zanglab.github.io/bart/>
6. BART3D: Binding Analysis for Regulation of Transcription for Three-Dimensional (3D) genomics data, a bioinformatics tool for identifying transcription regulators associated with genomic regions with differential chromatin interactions, given a pair of Hi-C datasets as input. <https://github.com/zanglab/bart3d>
7. BARTsc: Binding Analysis for Regulation of Transcription for single-cell (sc) omics data, a bioinformatics tool for identifying transcription regulators associated with cell clusters by integrating scRNA-seq, scATAC-seq, and scMultiome data. <https://github.com/zang-lab/BARTsc>
8. BART-spatial: Binding Analysis for Regulation of Transcription for spatial omics data, a bioinformatics tool for inferring transcriptional regulators whose activity are associated with a spatial pattern/trajectory from spatial omics (spatial transcriptomics or spatial epigenomics) data. <https://github.com/CorazonJW/BARTsp>
9. BARTweb: Interactive web server for performing BART analysis. <http://bartweb.org>
10. SICER2: An updated bioinformatics tool for SICER: Identification of broad enriched regions from ChIP-seq data. <https://zanglab.github.io/SICER2/>
11. GeneKnow: a large-language model-based AI tool for literature synthesis for gene-context analysis. <https://github.com/zang-lab/geneknow>
12. BART Cancer: A database resource for computationally predicted transcriptional regulator activities in 15 human cancers from The Cancer Genome Atlas (TCGA). <https://zanglab.github.io/bartcancer/>
13. CTCF explorer: an integrative data resource for CTCF binding in the human genome. <https://www.ctcf.info>
14. SELMA: Simplex Encoded Linear Model for Accessible chromatin, a computational model to accurately estimate and correct intrinsic enzymatic cleavage biases in chromatin accessibility profiling data. <https://github.com/zang-lab/SELMA>
15. PATTY: Pretrained Analyzer for Tn5 Transposase Yielded bias, a machine learning-based computational method for estimation and correction of Tn5-induced open-chromatin bias in CUT&Tag data. <https://github.com/zang-lab/PATTY>
16. IGN: Invariable Gene Normalization, a gene set based normalization method for chromatin accessibility profile data analysis. <https://github.com/zang-lab/IGN>

17. DORY: Differential chromatin tracing analysis, a computational method for identifying differential chromatin region pairs from image-based chromatin tracing data. <https://github.com/zang-lab/Dory>

HONORS AND AWARDS

- Eminent Scholar Award, International Association for Intelligent Biology and Medicine 2026
- Midcareer Faculty Fellow, Shannon Center for Advanced Studies, University of Virginia 2026
- Dean's Excellence in Faculty Research Award, University of Virginia School of Medicine 2024
- Vivian Pinn Scholar Award, University of Virginia School of Medicine 2024
- University of Virginia Research Award Honor Mention 2021
- University of Virginia Research Excellence Award 2020
(UVA junior faculty's highest research honor; only recipient from School of Medicine)
- MilliPub Club, University of Virginia School of Medicine 2018
- Seed Award, Jayne Koskinas Ted Giovanis Foundation for Health and Policy 2018
- Phi Beta Psi Sorority National Project Research Grant 2018
- Travel Award, Harvard Program in Quantitative Genomics 2014
- Leukemia and Lymphoma Society Fellow Award 2012–2015
- Dimitris N. Chorafas Foundation Prize 2010
- William Parke Prize in Theoretical Physics, The George Washington University 2009
- The Best Poster Award, the 7th Asia-Pacific Bioinformatics Conference 2009
- Chinese Government Award for Outstanding Self-financed Student Abroad 2008
- Student Travel Grant Award, Division of Biological Physics, American Physical Society 2008
- 2nd Award for Undergraduate Research, School of Physics, Peking University 2004
- Peking University Freshman Fellowship 2001
- 1st Prize, Beijing Region; 3rd Prize, National Final, Chinese Physics Olympiad 2000

RESEARCH SUPPORT

Current Research Support

1. NIH R35 GM133712 (MIRA) PI: Zang 09/21/2024–07/31/2029
Integrative computational models for functional epigenomics and transcriptional regulation
Role: **PI**
Effort: 45%
2. NIH R25 HG012918 MPI: Brown & Zang 06/01/2024–05/31/2027
Research Experiences in Genomic Applications of Responsible Data Science (REGARDS)
Role: **PI**
Effort: 4%
3. UVA Comprehensive Cancer Center PI: Zang 08/15/2025–08/14/2027
Characterizing epigenomic features of transcriptional condensates for cancer gene regulation
Role: **PI**

4. UVA School of Medicine Vivian Pinn Scholar Award PI: Zang 07/01/2024–06/30/2027
Data science methods for imaging-based spatial epigenomics
Role: **PI**

5. NIH R01 AI187349 PI: Xue 07/18/2025–06/30/2029
TLE transcription cofactors in anti-viral and anti-tumor immunity
Role: Subcontract PI
Effort: 5%

6. NIH R01 ES037626 PI: Cuddapah 07/01/2026–06/30/2030
Nickel exposure disrupts Sp1/ZEB1-dependent transcriptional regulation
Role: Subcontract PI
Effort: 5%

7. UVA Comprehensive Cancer Center PI: Zong 10/15/2025–10/14/2026
Determine the shifting transcriptional landscape in T cells during the transition from the initial
success to eventual failure of immune surveillance
Role: co-PI

8. NIH R01 AI189721 PI: Sun 04/01/2026–03/31/2031
Decoding cellular networks governing respiratory mucosal IgA immunity
Role: Co-Investigator
Effort: 2%

9. NIH R01 AG089953 PI: Smith 08/01/2025–05/31/2030
Mechanisms of rRNA instability during aging
Role: Co-Investigator
Effort: 5%

10. NIH R01 AI147394 PI: Sun 06/01/2025–05/31/2030
Host cellular network governing the development of post SARS-CoV-2 tissue sequelae
Role: Co-Investigator
Effort: 2.5%

11. NIH R01 AG090337 PI: Sun 12/01/2024–11/30/2029
CD38, T cells and post viral lung sequelae during aging
Role: Co-Investigator
Effort: 3%

12. NIH R01 CA178393 MPI: Loughran & Ratan 08/15/2024–07/31/2029
Genomic Architecture of LGL Leukemia
Role: Co-Investigator
Effort: 4%

13. NIH R01 HL173611 PI: Mohi 01/01/2025–11/30/2028
Molecular basis for progression of myeloproliferative neoplasms induced by JAK2V617F
Role: Co-Investigator
Effort: 3%

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|---|----------------|-----------------------|
| 14. NIH R01 HL148239 | PI: Miller | 07/01/2024–06/30/2028 |
| Multimodal genetic regulatory architecture of coronary artery disease | | |
| Role: Co-Investigator | | |
| Effort: 5% | | |
| 15. NIH R01 HL168611 | PI: Mohi | 06/15/2023–05/31/2027 |
| Molecular basis for myelodysplasia induced by U2AF1 mutations | | |
| Role: Co-Investigator | | |
| Effort: 2% | | |
| 16. NIH R01 CA259573 | PI: Stukenberg | 01/01/2022–12/31/2026 |
| Role of UTX condensation in chromatin regulation | | |
| Role: Co-Investigator | | |
| Effort: 4% | | |
| 17. Leukemia & Lymphoma Society (LLS) | PI: Tao | 09/01/2023–08/31/2027 |
| Mantle Cell Lymphoma Research Initiative | | |
| Role: Co-Investigator | | |
| Effort: 5% | | |

Completed Research Support

- | | | |
|--|-----------------|------------------------|
| 18. NIH R21 HG012981 | PI: Zang | 09/01/2023–08/31/2026 |
| A multi-level bias correction model for bulk and single-cell CUT&Tag data | | |
| Role: PI | | |
| Effort: 5% | | |
| 19. NIH R01 AI112579 | MPI: Xue & Zang | 05/01/2020–04/30/2025 |
| Tcf1 programs recall and reverses exhaustion in CD8 T cell response | | |
| Role: PI | | |
| Effort: 12% | | |
| 20. NIH R35 GM133712 (MIRA) | PI: Zang | 09/01/2019–09/20/2024 |
| Integrative computational models for functional epigenomics and transcriptional regulation | | |
| Role: PI | | |
| Effort: 51% | | |
| 21. NIH K22 CA204439 | PI: Zang | 09/01/2017–08/31/2019* |
| Quantitative modeling for chromatin regulation of gene expression in cancer | | |
| Role: PI | | |
| Effort: 75% | | |
| * Funded for 3 years but Year 03 fund was relinquished due to the funding of NIGMS MIRA (R35) award. | | |
| 22. Virginia Commonwealth Health Research Board | PI: Zang | 05/01/2021–07/31/2023 |
| Aberrant CTCF binding as an epigenetic signature of cancer | | |
| Role: PI | | |
| Effort: 10% | | |
| 23. Phi Beta Psi Charity Trust Research Grant | PI: Zang | 08/15/2018–08/14/2020 |
| Integrative modeling of multi-source genomic data for functional gene regulation in colorectal | | |

cancer
Role: **PI**

24. UVA Comprehensive Cancer Center PI: Zang 11/01/2023–10/31/2024
BARTsc: a computational method for transcription factor prediction from single-cell data
Role: **PI**

25. UVA Comprehensive Cancer Center PI: Zang 01/01/2023–12/31/2024
Roles of lncRNAs and R-loops in 3D genome organization and gene regulation in the cancer genome
Role: **PI**

26. UVA Comprehensive Cancer Center MPI: Wang,Li,Zang,Ratan 05/15/2024–05/14/2026
Deployment of the Swiss army knife in the human genome - CTCF licensing
Role: **PI**

27. NIH R21 CA274430 PI: Mohi 04/01/2023–03/31/2026
Targeting of IL-1 signaling in myelofibrosis
Role: Co-Investigator
Effort: 1%

28. NSF IIS-2106913 PI: Zhang 10/01/2021–09/30/2025
Knowledge-guided meta learning for multi-omics survival analysis
Role: Co-PI
Effort: 1%

29. NIH R21 CA283132 PI: Li 09/01/2023–08/31/2025
Racial disparities and colorectal DNA methylation-driven gene expression
Role: Co-Investigator
Effort: 3%

30. Chan Zuckerberg Initiative PI: Miller 07/01/2022–12/31/2024
MetaPlaq: Integrative single-cell meta-analysis for atherosclerosis
Role: Co-PI

31. NIH R01 ES031402 MPI: Cuddapah & Costa 01/01/2020–10/31/2024
Persistent transcriptional changes induced by nickel through epigenetic alterations
Role: Subcontract PI
Effort: 2.5%

32. NIH R01 NS116421 PI: Zhu 04/15/2021–06/30/2024
Developmental origin, injury and epigenomic regulation of NF1-associated peripheral nerve sheath tumors
Role: Subcontract PI
Effort: 4%

33. NIH R01 HL148239 PI: Miller 06/01/2019–04/30/2024
Cis-regulatory architecture of coronary vascular wall loci
Role: Co-Investigator
Effort: 5%

34. NIH R01 HL149893 PI: Mohi 01/01/2020–12/31/2023
The role of PTPN11 in myelofibrosis
Role: Co-Investigator
Effort: 1%
35. DOD/W81XWH-21-1-0008 PI: Mohi 01/01/2021–12/31/2023
Novel targeted therapy for metastatic triple-negative breast cancer
Role: Co-Investigator
Effort: 2%
36. NIH R01 GM132138 PI: Li 09/01/2019–07/31/2023
Genome-wide investigation of cis-splicing between adjacent genes
Role: Collaborator
Effort: 0% (because of Dr. Zang's funding of NIGMS R35 MIRA award)
37. NIH R01 NS115531 PI: Zhu 07/01/2019–06/30/2023
Investigating and targeting pathways of malignant peripheral nerve sheath tumor (MPNST)
Role: Subcontract PI
Effort: 2%
38. DOD/CDMRP Impact Award PI: Mohi 09/01/2019–06/30/2022
Targeted therapies for myeloproliferative neoplasms
Role: Co-Investigator
Effort: 1%
39. NIH U01 AG070960 PI: Yan 09/30/2020–05/31/2022
Machine learning-based multi-omics modeling and CRISPR/Cas9-mediated gene editing in elucidating molecular transducer of physical activity
Role: Co-Investigator
Effort: 5%
40. NIH R01 ES024727 PI: Cuddapah 02/01/2017–06/30/2020
Epigenetic reprogramming by nickel through chromatin domain disruption
Role: Subcontract PI
Effort: 8%
41. NFRI MPNST Pre-Clinical Research PI: Zhu 07/01/2018–01/31/2020
Involvement of PRC2 in malignant transformation and metastatic spread of NF1-associated MPNSTs
Role: Subcontract PI
Effort: 5%
42. Jayne Koskinas Ted Giovanis Foundation for Health and Policy Seed Award 8/1/2018–6/30/2019
Inferring lineage program regulating transcription factors for neuronal cell type identity from multi-modal single cell transcriptome and epigenome data
Role: Co-PI
43. American Cancer Society IRG 81-001-26 PI: Zang 12/18/2017–12/17/2018
Integrative modeling of functional gene regulation in colorectal cancer
Role: Project PI

44. Leukemia & Lymphoma Society Fellowship PI: Zang 07/01/2012–06/30/2015
 Computational studies on transcriptional regulation of Notch1 in T-ALL
 Role: PI
 Effort: 80%

TEACHING

2023 – now	Computational Biology Summer Institute (CGSI) University of California Los Angeles (UCLA)	Faculty
Spring 2017–	University of Virginia • COBI 8101/8102: Introduction to Computational Biology I/II • COBI 8301: Computational Genomics • BIMS 8601: Foundations of Computational Genomics • BIOC 8014: Chromatin II • BIOC 8145: Bioinformatics & Functional Analysis of Genomes • PHS 5705: Recent Advances in Public Health Genomics • BIMS 6000: Core Course in Integrative Biosciences • BME 1501: Introduction to Bioinformatics • BIMS 7100: Research Ethics • BIOL 4910: Undergraduate Research • CHEM 3951: Undergraduate Research	Co-instructor Co-instructor Co-instructor Co-instructor Co-instructor Co-instructor Co-instructor Guest Lecture Co-instructor Mentor Mentor
04/2021	New York University EHSC-GA-2050: Epigenetics and Environmental Diseases	Guest Lecture
05/2016	NIH/NCI Bioinformatics Workshop	Guest Lecture
11/2014	NIH/NCI Bioinformatics Workshop	Guest Lecture
09/2005–05/2008	The George Washington University Physics 001, Physics 002 Astronomy 001, Astronomy 002	Teaching Assistant Lab & Recitation Lab Instructor

INVITED TALKS

1. Department of Pathology, University of Michigan. Ann Arbor, MI. 11/13/2026
2. School of Life Sciences, Westlake University. Hangzhou, China. 10/13/2026
3. Department of Genetics, Rutgers the State University of New Jersey. Piscataway, NJ. 09/21/2026
4. Eminent Scholar Speech, 14th International Conference on Intelligent Biology and Medicine (ICIBM 2026). Buffalo, NY. 08/04/2026
5. Cancer Data Science Laboratory (CDSL) Seminar, Center for Cancer Research, National Cancer Institute, National Institutes of Health. Bethesda, MD. 05/06/2026
6. Department of Biochemistry and Molecular Biology Seminar, Johns Hopkins University. Baltimore, MD. 03/25/2026
7. Department of Biology Seminar, Texas A&M University. College Station, TX. 01/20/2026

8. School of Life Sciences, Tsinghua University. Beijing, China. 12/26/2025
9. 13th International Chinese Statistical Association (ICSA) International Conference. Taipei, Taiwan. 12/18/2025
10. Cold Spring Harbor Asia Meeting: Systems Biology of Gene Regulation and Genome Editing. Suzhou, China. 10/21/2025
11. Cell Systems and Anatomy Seminar Series, University of Texas Health Science Center San Antonio. San Antonio, TX. 09/30/2025
12. Center for Computational Biology and Bioinformatics Seminar Series, Indiana University School of Medicine. Indianapolis, IN. 09/12/2025
13. Computational Genomics Summer Institute (CGSI), University of California Los Angeles. Los Angeles, CA. 07/14/2025
14. University Program in Genetics and Genomics Seminar Series, Duke University. Durham, NC. 01/21/2025
15. Keynote talk, Virginia Soft Matter Workshop, James Madison University. Harrisonburg, VA. 10/18/2024
16. Joint Statistical Meetings (JSM). Portland, OR. 08/06/2024
17. Computational Genomics Summer Institute (CGSI), University of California Los Angeles. Los Angeles, CA. 07/17/2024
18. Department Seminar Series, Center for Genetic Medicine, Children's National Hospital. Washington, DC. 06/21/2024
19. Frontiers in Bioinformatics Seminar Series, University of California Los Angeles. Los Angeles, CA. 04/15/2024
20. MidSouth Computational Biology and Bioinformatics Society (MCBIOS) Annual Conference. Atlanta, GA. 03/23/2024
21. American Physical Society (APS) March Meeting. Minneapolis, MN. 03/07/2024
22. Faculty Research Retreat, University of Virginia School of Medicine. Charlottesville, VA. 02/03/2024
23. Center for Statistical Science, Tsinghua University. Beijing, China. 10/30/2023
24. Cold Spring Harbor Asia Meeting: Systems Biology of Gene Regulation and Genome Editing. Suzhou, China. 10/27/2023
25. Seminar Series, The Jackson Laboratory for Genomic Medicine, Farmington, CT. 10/17/2023
26. School of Data Science Research Retreat, University of Virginia. Charlottesville, VA. 09/22/2023
27. Department of Microbiology, Tumor and Cell Biology, Karolinska Institutet. Stockholm, Sweden. 08/02/2023
28. Computational Genomics Summer Institute (CGSI), University of California Los Angeles. Los Angeles, CA. 07/20/2023
29. Distinguished Lectures in Genetics Seminar Series, Cincinnati Children's Hospital Medical Center, Cincinnati, OH. 06/27/2023
30. Gordon Research Conference: Cancer Genetics and Epigenetics. Barga, Italy. 05/25/2023
31. American Association for Cancer Research (AACR) Annual Meeting. Session ED045: Distilling data from single-cell and spatial RNA-seq: a new perspective, a new approach. Orlando, FL. 04/15/2023
32. Carter Immunology Center, University of Virginia. Charlottesville, VA. 02/20/2023
33. Oncological Data Science Seminar Series, Moffitt Cancer Center. Tampa, FL. 09/20/2022 (virtual)
34. International Conference on Intelligent Biology and Medicine (ICIBM 2022). Philadelphia, PA.

08/08/2022

35. Computational Biology Program, Boston Children's Hospital, Harvard Medical School. Boston, MA. 07/29/2022
36. Department of Computational Biomedicine, Cedars-Sinai Medical Center, Los Angeles, CA. 05/31/2022 (virtual)
37. Department of Biology, Johns Hopkins University. Baltimore, MD. 12/03/2021 (virtual)
38. Division of Biostatistics and Bioinformatics, Department of Epidemiology and Public Health and Department of Neurosurgery, University of Maryland School of Medicine. Baltimore, MD. 06/03/2021
39. Department of Environmental Medicine, New York University. New York, NY. 05/11/2021 (virtual)
40. Child Health Research Center, University of Virginia. Charlottesville, VA. 04/23/2021 (virtual)
41. Foundation of Biomedical Data Science Seminar Series, University of Virginia School of Data Science. Charlottesville, VA. 04/02/2021 (virtual)
42. Computational Biology and Bioinformatics Seminar Series, Duke University. Durham, NC. 03/29/2021 (virtual)
43. Center for Cancer Genomics, Department of Biochemistry and Molecular Genetics, Northwestern University. Chicago, IL. 12/09/2020 (virtual)
44. 14th International Bioinformatics Workshop (IBW 2019), Beijing, China. 08/03/2019
45. State Key Laboratory of Cardiovascular Disease, Peking Union Medical College. Beijing, China. 07/31/2019
46. School of Mathematics, Shandong University. Jinan, China. 07/29/2019
47. Department of Cognitive Science, University of California, San Diego, CA. 12/20/2018
48. Biomedical Pioneering Innovation Center, Peking University. Beijing, China. 08/21/2018
49. Department of Pathology, University of Virginia. Charlottesville, VA. 12/19/2017
50. Department of Biostatistics, Virginia Commonwealth University. Richmond, VA. 11/3/2017
51. Center for Quantitative Biology, Peking University. Beijing, China. 09/29/2017
52. Institute of Biophysics, Chinese Academy of Sciences. Beijing, China. 09/28/2017
53. Beijing Institute for Scientific and Engineering Computing, Beijing University of Technology. Beijing, China. 09/27/2017
54. Center for Statistical Science, Tsinghua University. Beijing, China. 09/25/2017
55. Department of Physics, Fudan University. Shanghai, China. 09/23/2017
56. CAS-MPG Partner Institute for Computational Biology, Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences. Shanghai, China. 09/18/2017
57. School of Life Sciences, Peking University. Beijing, China. 06/08/2016
58. Workshop on Analysis of ChIP-Seq Data, National Cancer Institute, National Institutes of Health. Bethesda, MD. 05/17/2016
59. Laboratory of Genome Integrity, National Cancer Institute, National Institutes of Health. Bethesda, MD. 04/07/2016
60. Department of Immunobiology, Yale University. New Haven, CT. 03/11/2016
61. University of Virginia. Charlottesville, VA. 02/05/2016
62. The Wistar Institute. Philadelphia, PA. 01/29/2016
63. Academy of Mathematics and Systems Science, Chinese Academy of Sciences. Beijing, China.

01/07/2016

64. Peking University Third Hospital. Beijing, China. 01/05/2016
65. Peking University Medical Center. Beijing, China. 01/04/2016
66. University of Illinois College of Medicine, Peoria, IL. 12/17/2015
67. The Stadtman Symposium, National Institutes of Health. Bethesda, MD. 12/10/2015
68. National Heart, Lung, and Blood Institute, National Institutes of Health. Bethesda, MD. 11/20/2015
69. University of Nevada School of Medicine, Reno, NV. 10/06/2015
70. College of Pharmacy, University of Rhode Island. Kingston, RI. 06/24/2015
71. Uniformed Services University of the Health Sciences. Bethesda, MD. 05/29/2015
72. Department of Electrical & Computer Engineering, University of Nebraska, Lincoln, NE. 02/18/2015
73. National Human Genome Research Institute, National Institutes of Health. Bethesda, MD. 02/04/2015
74. National Cancer Institute, National Institutes of Health. Bethesda, MD. 12/19/2014
75. Cancer epigenetics symposium, Harvard Medical School. Boston, MA. 12/14/2014
76. Biophysics Seminar, The George Washington University. Washington, DC. 10/02/2014
77. Workshop on big data analysis in biomedical research, Harvard Medical School. Boston, MA. 06/24/2014
78. CAS-MPG Partner Institute for Computational Biology, Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences. Shanghai, China. 11/28/2013
79. Institute of Systems Biomedicine, Shanghai Jiao Tong University. Shanghai, China. 11/26/2013
80. Computational Biology and Bioinformatics Organization, Broad Institute. Cambridge, MA. 10/16/2013
81. Megakaryocyte and Platelet Biology Joint Meeting, Boston Children's Hospital. Boston, MA. 09/12/2012
82. Department of Animal and Avian Sciences, College of Agriculture and Natural Resources, University of Maryland, College Park, MD. 09/28/2009

CONTRIBUTED PRESENTATIONS

1. Gordon Research Conference on Single-Cell Genomics. Les Diablerets, Switzerland, 05/2026. (*Poster + 1 Poster by a trainee*)
2. American Association for Cancer Research (AACR) Annual Meeting. San Diego, CA, 04/2026. (*Poster + 1 Poster by a trainee*)
3. American Physical Society (APS) Global Physics Summit. Denver, CO, 03/2026. (*Talk*)
4. Cold Spring Harbor Meeting on Systems Biology: Global Regulation of Gene Expression, 03/2026 (*Poster + 1 Poster by a trainee*)
5. Advances in Genome Biology and Technology (AGBT) General Meeting. Orlando, FL, 02/2026. (*Poster*)
6. Cold Spring Harbor Meeting on Mechanisms of Eukaryotic Transcription, 08/2025. (*Poster + 1 Poster by a trainee*)
7. American Association for Cancer Research (AACR) Annual Meeting. Chicago, IL, 04/2025. (*4 Posters presented by trainees*)

8. American Physical Society (APS) Global Physics Summit. Anaheim, CA, 03/2025. (*Talk*)
9. American Association for Cancer Research (AACR) Special Conference: Functional and Genomic Precision Medicine in Cancer. Boston, MA, 03/2025. (*Poster*)
10. Advances in Genome Biology and Technology (AGBT) General Meeting. Marco Island, FL, 02/2025. (*Poster*)
11. Gordon Research Conference on Single-Cell Genomics. Les Diablerets, Switzerland, 05/2024. (*Poster*)
12. American Association for Cancer Research (AACR) Annual Meeting. San Diego, CA, 04/2024. (*Poster + 4 Posters presented by trainees*)
13. Cold Spring Harbor Meeting on Systems Biology: Global Regulation of Gene Expression, 03/2024 (*Poster + 1 Poster by a trainee*)
14. American Physical Society (APS) March Meeting. Minneapolis, MN, 03/2024. (*Talk*)
15. Chan Zuckerberg Science Initiative Single-Cell Biology Annual Meeting. San Diego, CA, 11/2023. (*Talk*)
16. ISMB/ECCB 2023: 31th Conference on Intelligent Systems for Molecular Biology. Lyon, France. 07/2023. (*Poster*)
17. American Association for Cancer Research (AACR) Annual Meeting. Orlando, FL, 04/2023. (3 *Posters presented by trainees*)
18. Keystone Meeting on Epigenetics, Chromatin, Development and Disease. Victoria, BC, Canada, 03/2023. (*Poster*)
19. American Physical Society (APS) March Meeting. Las Vegas, LV, 03/2023. (*Talk*)
20. Chan Zuckerberg Science Initiative Single-Cell Biology Annual Meeting. San Jose, CA, 11/2022. (*Talk*)
21. ISMB 2022: 30th Conference on Intelligent Systems for Molecular Biology. Madison, WI, 07/2022. (*Poster*)
22. Gordon Research Conference on Single-Cell Cancer Biology. Easton, MA, 06/2022. (3 *Posters presented by trainees*)
23. Cold Spring Harbor Meeting on Systems Biology: Global Regulation of Gene Expression, 03/2022. (3 *Posters presented by trainees*)
24. Cold Spring Harbor Meeting on Mechanisms of Eukaryotic Transcription, 09/2021. (*Poster presented by a trainee*)
25. Cold Spring Harbor Meeting on Eukaryotic DNA Replication & Genome Maintenance, 09/2021. (*Poster presented by a trainee*)
26. 2021 NHLBI Systems Biology Symposium (Virtual), 05/2021. (5 *Posters presented by trainees*)
27. American Association for Cancer Research (AACR) Annual Meeting (Virtual), 04/2021. (*Poster*)
28. American Physical Society (APS) March Meeting (Virtual), 03/2021. (*Talk*)
29. American Association for Cancer Research (AACR) Annual Meeting (Virtual), 06/2020. (*Poster*)
30. iPoLS 2020: International Physics of Living Systems Annual Meeting (Virtual), 06/2020. (*Poster*)
31. Cold Spring Harbor Meeting on Systems Biology: Global Regulation of Gene Expression (Virtual), 03/2020. (*Poster presented by a trainee*)
32. Keystone Meeting on Cancer Epigenetics. Keystone, CO, 01/2020. (*Poster*)
33. Physics Meets Biology 2019, Institute of Physics, University of Oxford. Oxford, UK, 09/2019. (*Poster*)

34. RECOMB 2019: 23rd Annual International Conference on Research in Computational Molecular Biology. Washington, DC, 05/2019 (*5 Posters presented by trainees*)
35. Gordon Research Conference on Cancer Genetics and Epigenetics. Lucca, Italy, 04/2019. (*Poster*)
36. American Association for Cancer Research (AACR) Annual Meeting. Atlanta, GA, 04/2019. (*Poster*)
37. American Physical Society (APS) March Meeting. Boston, MA, 03/2019. (*Talk*)
38. ISMB 2018: Conference on Intelligent Systems for Molecular Biology. Chicago, IL, 07/2018. (*Poster*)
39. Biophysics of Epigenetic and Chromatin Dynamics Workshop, Higgs Centre for Theoretical Physics, University of Edinburgh. Edinburgh, UK, 04/2018. (*Poster*)
40. American Physical Society (APS) March Meeting. Los Angeles, CA, 03/2018. (*Talk*)
41. Keystone Meeting on Cancer Epigenetics. Breckenridge, CO, 02/2018. (*Poster*)
42. Cold Spring Harbor Asia Meeting on Precision Cancer Biology. Suzhou, China, 09/2017. (*Poster*)
43. Keystone Meeting on Epigenetics and Human Disease. Seattle, WA, 01/2017. (*Poster*)
44. NCI Symposium on Chromosome Biology. Bethesda, MD, 11/2016. (*Poster*)
45. ENCODE Consortium Meeting. San Diego, CA, 06/2016. (*Poster*)
46. AACR Special Conference on Chromatin and Epigenetics in Cancer. Atlanta, GA, 09/2015. (*Poster*)
47. Keystone Meeting on Cancer Epigenetics. Santa Fe, NM, 02/2014. (*Poster*)
48. NCI Symposium on Chromosome Biology. Bethesda, MD, 04/2013. (*Poster*)
49. Cold Spring Harbor Laboratory Meeting on Systems Biology: Global Regulation of Gene Expression. Cold Spring Harbor, NY, 03/2012. (*Poster*)
50. Cold Spring Harbor Laboratory Meeting on Systems Biology: Global Regulation of Gene Expression. Cold Spring Harbor, NY, 03/2010. (*Poster*)
51. The 7th Asia Pacific Bioinformatics Conference. Beijing, China, 01/2009. (*Best Poster Award Winner*)
52. American Physical Society (APS) March Meeting. New Orleans, LA, 03/2008. (*Talk*)

MENTEES AND STUDENTS

Postdoctoral Associates Mentored (6 completed, 2 ongoing)

1. Zhenjia Wang, PhD (UVA, 01/2017–01/2024)
2. Sheng'en Shawn Hu, PhD (UVA, 11/2017–11/2021, now Res. Asst. Prof. at UVA)
3. Christopher Dampier, MD (UVA, joint with Graham Casey, 07/2019–05/2021)
4. Qingying Chen, PhD (UVA, 01/2021–06/2024)
5. William F. Lamberti, PhD (UVA, 01/2021–08/2022, now Asst. Prof. at GMU)
6. Yidan Cong, PhD (UVA, 06/2021–02/2022)
7. Shengyuan Wang, PhD (UVA, 03/2022–)
8. Ze Yan, PhD (07/2025–)

Graduate Students Mentored (4 masters, 4 PhD rotation, 1 PhD graduated; 6 PhD ongoing)

1. Wuwei Tan (MS in Statistics, UVA, 03/2017–05/2018)
2. Yushan Brandon Feng (MS in Statistics, UVA, 05/2018–05/2019, now Asst. Prof. at Univ. Texas Austin)

3. Seung Hyun Lee (MS in Statistics, UVA, 09/2018–05/2019)
4. Yifan Zhang (Biomedical Engineering, UVA, 08/2018–12/2019)
5. Audrey Putelo (PhD in Biomedical Sciences, UVA, Rotation 07/2020–08/2020)
6. Jose Verdezoto (PhD in Biomedical Sciences, UVA, Rotation 10/2020–11/2020)
7. Jamel Simpson (PhD in Biomedical Sciences, UVA, Rotation 1/2021–2/2021)
8. Hongpan Zhang (PhD in Biomedical Sciences, UVA, 07/2021–08/2026)
9. Mengxue Tian (PhD in Biomedical Sciences, UVA, 07/2021–)
10. Zhaoxia Ma (PhD in Biomedical Sciences, UVA, 09/2021–)
11. Jingyi Wang (PhD in Biomedical Sciences, UVA, 07/2024–)
12. Bhummanat Sittipongpittaya (PhD in Biomedical Sciences, UVA, 1/2025–)
13. Sophie O’Keefe (PhD in Biomedical Sciences, UVA, Rotation 1/2025–2/2025)
14. Danielle Hughes (PhD in Biomedical Sciences, UVA, joint with Thomas Laughran, 7/2025–)
15. Patrick Hart (PhD in Experimental Pathology, joint with Clint Miller, 8/2026–)

PhD Thesis Committee Service (16 completed, 3 ongoing)

University of Virginia:

1. Jeffrey Xing (PhD in Biomedical Engineering, 01/2017–10/2020)
2. Yang Yu (PhD in Biology, 12/2020–10/2022)
3. Jacob Wolpe (PhD in Cell Biology, 07/2021–06/2023)
4. Zhijian He (PhD in Biomedical Engineering, 02/2023–04/2024)
5. Justin Elfman (PhD in Biochemistry and Molecular Genetics, 06/2019–07/2024)
6. Yuanming Liu (PhD in Biology, 01/2021–06/2025)
7. Nicole Kirk (PhD in Microbiology, Immunology, and Cancer Biology, 12/2021–08/2025)
8. Samuel Haddox (PhD in Biochemistry and Molecular Genetics, 02/2021–11/2025, Committee Chair)
9. Yipei Song (PhD in Electrical and Computer Engineering, 04/2024–12/2025)
10. Louisa Quesada (PhD in Biomedical Engineering, 11/2022–03/2026)
11. Haowen Zhou (PhD in Statistics, 12/2025–04/2026)
12. Jose Verdezoto (PhD in Biochemistry and Molecular Genetics, 05/2022–06/2026, Committee Chair)
13. Patrick Hart (PhD in Experimental Pathology, 05/2026–)
14. Zishan Peng (PhD in Biochemistry and Molecular Genetics, 07/2026–)

External committee service:

15. Celestia Fang (PhD in Biochemistry and Molecular Genetics, Northwestern University Feinberg School of Medicine, 09/2018–06/2021)
16. Evan Tarbell (PhD in Biochemistry, University at Buffalo, 11/2018–12/2018)
17. Ruoyun Wang (PhD in Molecular & Cell Biology, Dartmouth College, 6/2024)
18. Christy Lee (PhD in Statistics, UCLA, 12/2023–06/2025)
19. Lian Duan (PhD in Biostatistics, University of Illinois Chicago, 07/2026–)

Undergraduate Student Research Supervised (32 completed, 1 ongoing)

1. Celina Shih (Johns Hopkins University, 05/2017–07/2017)
2. Yayi Feng (UVA, 09/2017–05/2020)
3. Stephany Perez-Rojas (UVA, 01/2018–05/2020)
4. John Zhang (UVA, 02/2018–05/2020)
5. Yuantao Chen (UVA, 02/2018–05/2021)

6. Jeffrey Yoo (UVA, 02/2018–05/2020)
7. Naifei Pan (UVA, 04/2018–02/2019)
8. Alexander An (UVA, 05/2018–08/2018)
9. Zhou Lu (UVA, 08/2018–05/2020)
10. Allison Ryu (UVA, 09/2018–12/2018)
11. Aaron Ho (UVA, 10/2018–05/2020)
12. Benjamin Ke (UVA, 01/2019–07/2022)
13. Sophia Park (UVA, 01/2019–08/2019)
14. Tianhe Li (UVA, 01/2019–05/2020)
15. Leyao Li (UVA, 03/2019–05/2020)
16. Megan Grieco (UVA, 04/2019–07/2022)
17. Xinyue Qiu (UVA, 04/2019–05/2021)
18. Zack Thomas (UVA, 04/2019–05/2021)
19. Sekou Rowe (NSF REU student, Howard University, 05/2019–08/2019)
20. Minaal Khan (UVA, 09/2019–08/2021)
21. Kendall Davis (UVA, 10/2019–08/2021)
22. Anvitha Chimata (UVA, 10/2019–08/2021)
23. Andrew Vanichkachorn (UVA, 10/2019–07/2022)
24. Jingwen Wang (UVA, 01/2020–05/2021)
25. Emilie Neulen (UVA, 02/2020–08/2022)
26. Kuki Davey (UVA, 07/2020–05/2022)
27. Alex Hickman (UVA, 07/2020–05/2021)
28. Jingyi Wang (UVA, 07/2022–05/2024)
29. Ibrahim Khanani (UVA SRIP program, Duke University, 05/2024–08/2024)
30. Andy Cao (UVA, 08/2022–05/2025)
31. Yanbin Xu (UVA, 10/2022–05/2025)
32. Grant Atwood (UVA, Engineering School Summer Research Fellow, 08/2023–05/2026)
33. Katherine Zhang (UVA, 08/2024–)

Visiting Students/Scholars Supervised (7 completed)

1. Yiren Wang (Graduate from Peking University, 07/2017–06/2018)
2. Qi Li (PhD Candidate in Statistics, Tsinghua University, 01/2018–02/2018, 03/2019)
3. Wen Shen (Undergraduate Student, Nanjing University, 02/2018–05/2018)
4. Rongquan He, MD (PhD Candidate, Guangxi Medical University, 03/2018–02/2019)
5. Wenjing Ma (Graduate from Beijing University of Posts and Telecommunications, 08/2018–06/2019, now Asst. Prof. at Chinese University of Hong Kong)
6. Gang Chen, MD, PhD (Professor of Pathology, Guangxi Medical University, 09/2018–08/2019)
7. Ruoyao Shi (Undergraduate Student, Jilin University, 10/2019–05/2020)

High School Students Supervised (5 completed, 1 ongoing)

1. Katherine Zhang (Thomas Jefferson High School for Science and Technology, Class of 2024, 06/2022–08/2022)
2. Elizabeth McClung (Rockbridge County High School, Class of 2023, UVA Advance Summer Program, 07/2022–08/2022)
3. Riya Shah (McLean High School, Class of 2023, UVA Advance Summer Program, 07/2022–08/2022)
4. Chris Stewart (Monticello High School, Class of 2023, UVA Advance Summer Program, 07/2022–08/2022)
5. Ally Kuang (Bob Jones High School, Class of 2026, 07/2025–08/2025)

6. Alexander Chen (Albemarle High School, Class of 2027, 07/2025–)

PROFESSIONAL SERVICE

Grant Review/Study Section Service

Standing Member,
Ad hoc **Co-Chair** National Institutes of Health (NIH) Center for Scientific Review
Maximizing Investigators' Research Award (MRAF) Study Section (2023–
2026)

Ad hoc grant reviewer

- Austrian Science Fund (FWF) (2026)
- Israel Science Foundation (2026)
- Dutch Cancer Society (KWF Kankerbestrijding) (2026)
- Swiss National Science Foundation (SNSF) (2025)
- Canada Research Chairs (2025)
- Wellcome Trust (UK) external grant (2023)
- UK Research and Innovation (UKRI):
 - Biotechnology and Biological Sciences Research Council (BBSRC, UKRI) (2022)
 - Medical Research Council (MRC, RCUK) (2017)
- US National Institutes of Health (NIH) Center for Scientific Review:
 - **Chair**, ZCA1 SRB-P (M1) Cancer Target Discovery and Development Network (CTD2) Special Emphasis Panel (03/2023)
 - Member, ZRG1 MGG-C (55) Maximizing Investigators' Research Award (MRAA) Special Emphasis Panel (11/2022)
 - Member, ZRG1 GGG-S (55): Maximizing Investigators' Research Award (MIRA) for Early Stage Investigators (ESI) Special Emphasis Panel (03/2022)
 - Member, ZRG1 CB-K (55): Maximizing Investigators' Research Award (MIRA) for Early Stage Investigators (ESI) Special Emphasis Panel (03/2021)
 - Member, ZMH1 ERB-M (06): BRAIN Initiative Cell Census Network (BICCN) Special Emphasis Panel (07/2020)
 - Member, ZRG1 PSE-B (04): Epidemiology and Statistical Methods Special Emphasis Panel (03/2020)
- Netherlands Organisation for Scientific Research (NWO) (2017)
- UVA internal review committee for Virginia CHRB awards (2021)
- UVA Carter Immunology Center internal review committee (2021)
- Inova Translational Research Funding Program (2018)

Journal Editorship Service / Peer Review

Associate Editor *Science Advances* (2026–)

Editorial Board Member *Epigenomes* (2026–)

Scientific Editor *PLOS Computational Biology* (2024–)

<i>Advisory Board Member</i>	<i>Patterns (Cell Press) (2024–)</i>
<i>Editorial Board Member</i>	<i>Immunoinformatics (2022–2026)</i>
<i>Topical Advisory Board</i>	<i>Epigenomes (2021–2026)</i>
<i>Guest Editor</i>	<i>PLOS Computational Biology (2020–2024)</i>
<i>Ad hoc journal reviewer</i>	<i>Cell</i> <i>Nature Biotechnology</i> <i>Nature Genetics</i> <i>Nature Methods</i> <i>Nature Cancer</i> <i>Nature Communications</i> <i>Science Advances</i> <i>Cell Genomics</i> <i>Cell Systems</i> <i>Genome Research</i> <i>Genome Biology</i> <i>Cancer Research</i> <i>EMBO Journal</i> <i>Nucleic Acids Research</i> <i>Cellular and Molecular Life Sciences</i> <i>Bioinformatics</i> <i>Briefings in Bioinformatics</i> <i>Genomics, Proteomics & Bioinformatics</i> <i>Epigenetics</i> <i>PLOS Computational Biology</i> <i>BMC Bioinformatics</i> <i>BMC Genomics</i> <i>Breast Cancer Research</i> <i>Journal of the American Statistical Association</i> <i>Advanced Science</i> <i>Scientific Reports</i> <i>iScience</i> <i>Cell Discovery</i> <i>Communications Biology</i> <i>Open Biology</i> <i>Patterns</i> <i>PLOS One</i> <i>NAR Cancer</i> <i>Journal of Computational Biology</i> <i>Quantitative Biology</i> <i>Journal of Theoretical Biology</i> <i>IEEE/ACM Transactions on Computational Biology & Bioinformatics</i> <i>Computational Statistics and Data Analysis</i> <i>Biomolecules</i> <i>Frontiers of Medicine</i> <i>Frontiers in Oncology</i>

National/Extramural Community Service

<i>Faculty Mentor</i>	Biomedical Data Science Innovation Lab: Quantitative Approaches in Spatial Multi-Omics for Guiding Personalized Medicine, 2025
<i>Co-Chair</i>	American Physical Society (APS) March Meeting Sessions on Genome Organization and Subnuclear Phenomena, 2024–
<i>Faculty Member</i>	Computational Genomics Summer Institute (CGSI), an international summer school hosted by University of California Los Angeles (UCLA), 2023–
<i>Chair</i>	NIH CSR Special Emphasis Panel ZCA1 SRB-P (M1): Cancer Target Discovery and Development Network (CTD2), 2023
<i>Member</i>	Scientific Advisory Board (SAB), Integrated Translational Health Research Institute of Virginia (iTHRIV), 2023–2025
<i>Faculty Mentor</i>	Biomedical Data Science Innovation Lab: Challenges in Brain Analytics and Integration, 2020–2021
<i>Co-Chair</i>	American Association for Cancer Research (AACR) Annual Meeting Minisymposium: Transcriptional Gene Regulation in Cancer, 2020

University Service/Leadership

<i>Director of Computational Genomics</i>	UVA Comprehensive Cancer Center, 7/2024–
<i>Member</i>	Faculty Promotion and Tenure Committee, UVA Department of Genome Sciences, 7/2026–
<i>Member</i>	Executive Committee, UVA Department of Genome Sciences, 4/2026–
<i>Member</i>	Faculty Advisory Committee, UVA School of Medicine Bioinformatics Core Facility, 11/2025–
<i>Member</i>	AI Research Committee, UVA Comprehensive Cancer Center, 4/2025–
<i>Faculty Advisor</i>	Data Science Shared Resource, UVA Comprehensive Cancer Center, 2/2025–
<i>Member</i>	Faculty Advisory Committee for AI Research, UVA Health, 2/2025–
<i>Chair</i>	Faculty Search Committee, UVA Precision Medicine/Precision Health Initiative, Department of Genome Sciences and UVA Comprehensive Cancer Center, 2024–2025
<i>Member</i>	Steering Committee, PhD Program in Computational Biology, UVA School of Medicine, 8/2024–

Co-Chair UVA Comprehensive Cancer Center Molecular Epidemiology Faculty Search Committee, 2022

Chair Emerging Scholars in Genome Sciences Symposium Planning Committee
(*Member*, 2020– ; *Chair*, 2021–2024)

Updated: September 9, 2026